Fundamentals of Probability

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Probability

- How likely something is to happen.
- Many events can't be predicted with total certainty. The best we can say is how likely they are to happen, using the idea of probability.
- Tossing a Coin

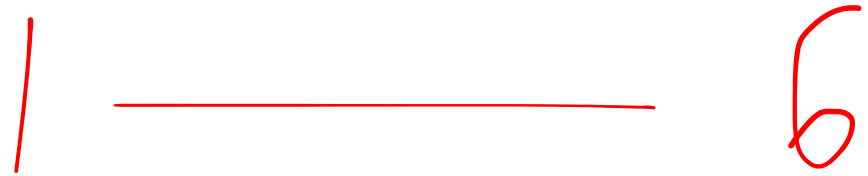


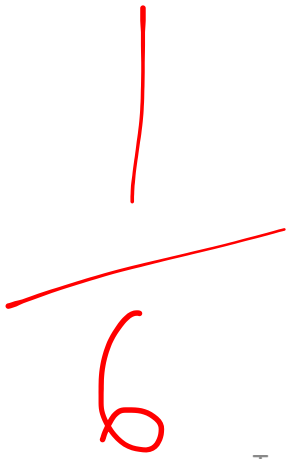
H / T = 1/2

Probability

- Throwing Dice



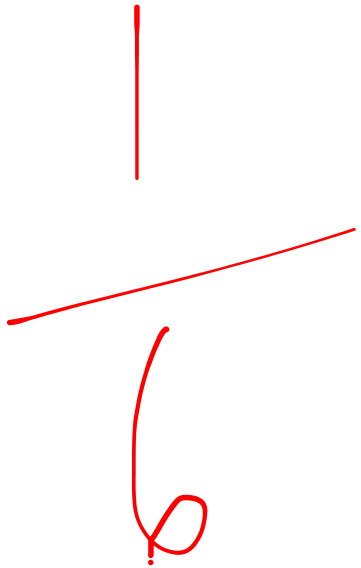




Probability Formula

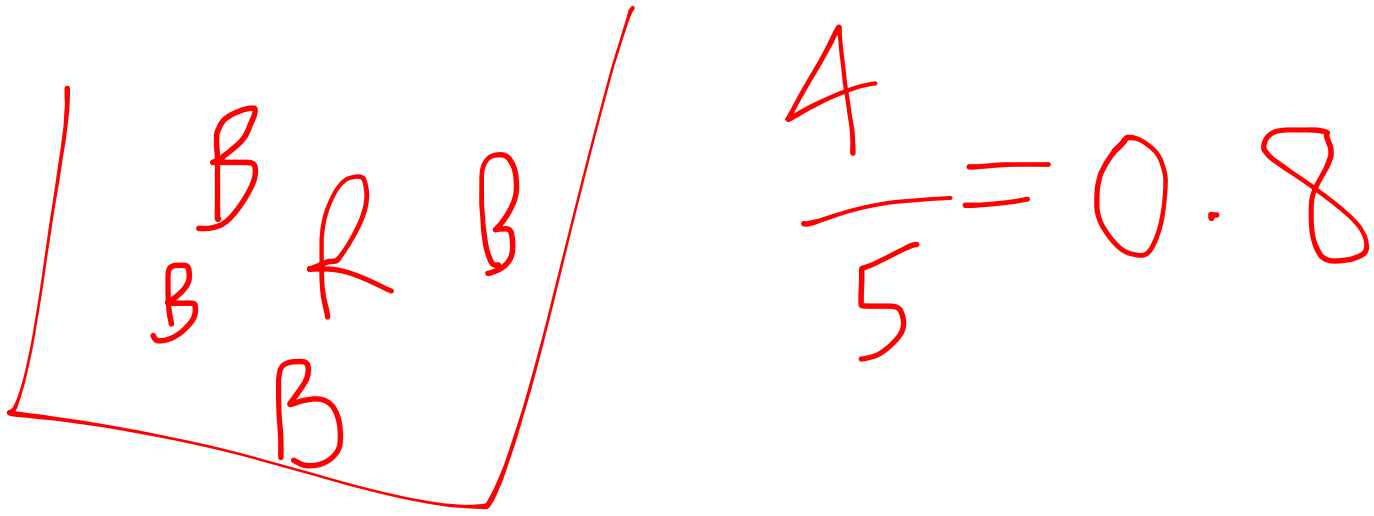
$$\text{Probability of an event happening} = \frac{\text{Number of ways it can happen}}{\text{Total number of outcomes}}$$

Example: the chances of rolling a "4" with a die



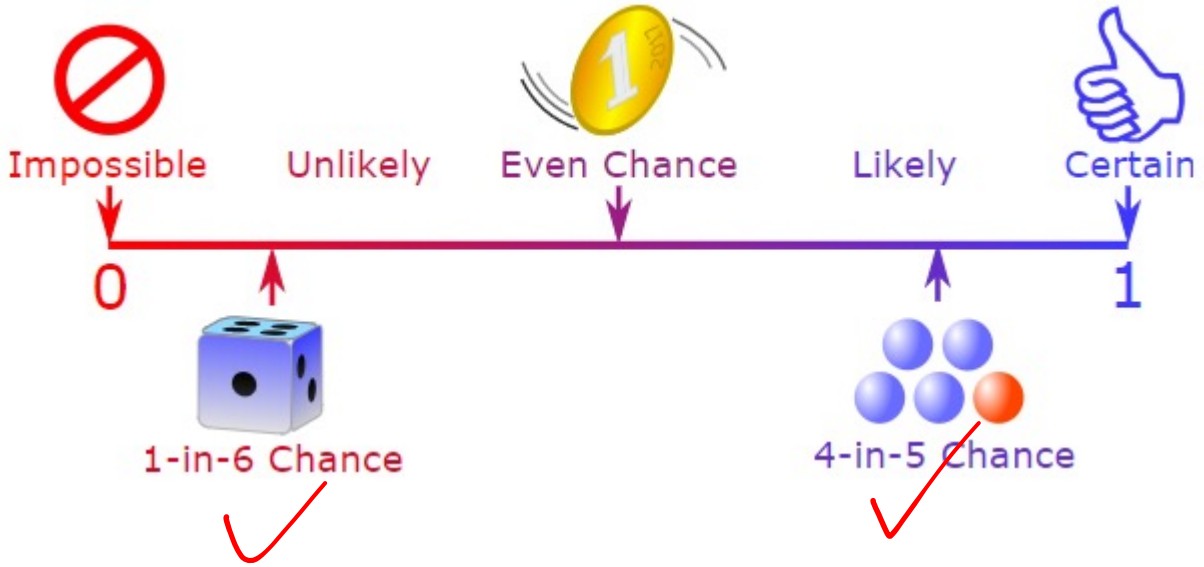
Example:

- There are 5 marbles in a bag: 4 are blue, and 1 is red. What is the probability that a blue marble gets picked?



Probability Line

- Probability is the **chance** that something will happen. It can be shown on a line:



Range of the probability

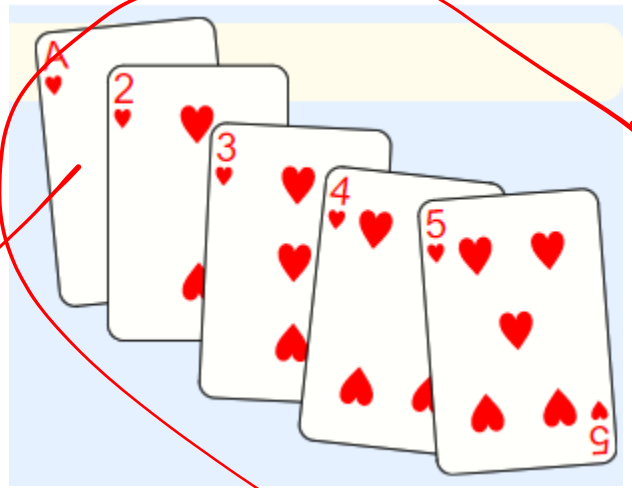
- **Between 0 and 1**

- The probability of an event will **not** be less than 0.

- The probability of an event will **not** be more than 1.

Special terminologies in Probability

- **Sample Space:** all the possible outcomes of an experiment.
- Example: choosing a card from a deck



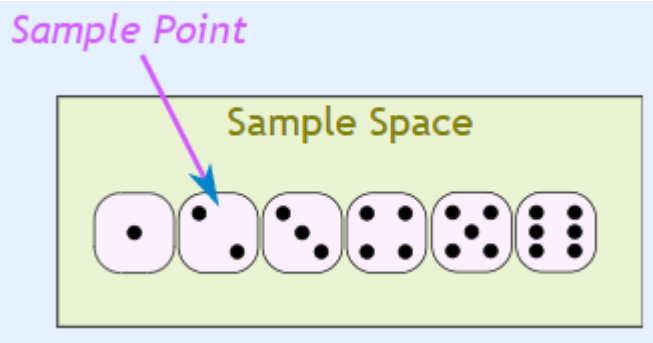
S.P

Special terminologies in Probability

- The Sample Space is made up of Sample Points:
- **Sample Point:** just one of the possible outcomes
- **Example:** Deck of Cards
- the 5 of Clubs is a sample point
- the King of Hearts is a sample point 
- "King" is not a sample point. There are 4 Kings, so that is 4 *different* sample points.

Special terminologies in Probability

- The Sample Space is made up of Sample Points:
- **Sample Point:** just one of the possible outcomes
- Example: Throwing dice
- There are 6 different sample points in the sample space.





Basic Counting Principle

- When there are **m** ways to do one thing, and **n** ways to do another,
  then there are **m×n** ways of doing **both**.
- Example: you have **3** shirts and **4** pants.

12 outfits

- Example: There are **6** flavors of ice-cream, and **3** different cones.

18

Basic Counting Principle

- It also works when you have more than 2 choices:
- Example: You are buying a new car.

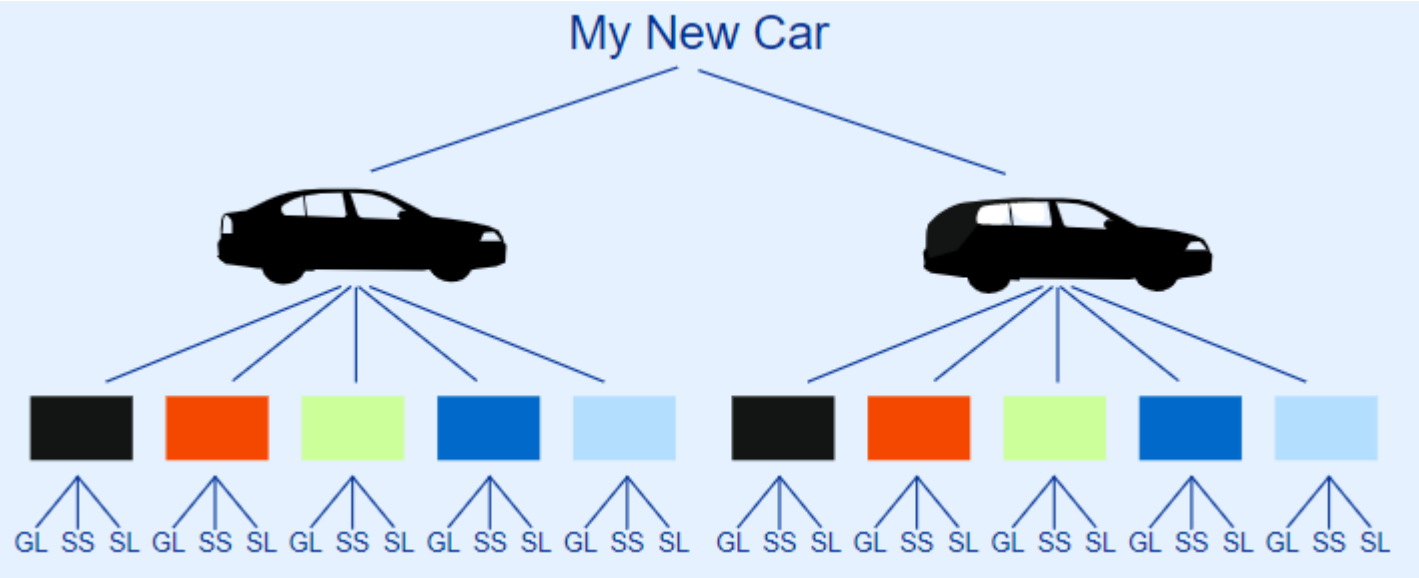


- There are 5 colors available:



- There are 3 models:
  - GL (standard model),
  - SS (sports model with bigger engine)
  - SL (luxury model with leather seats)
- How many total choices?
- You can see in this "tree" diagram:

Basic Counting Principle

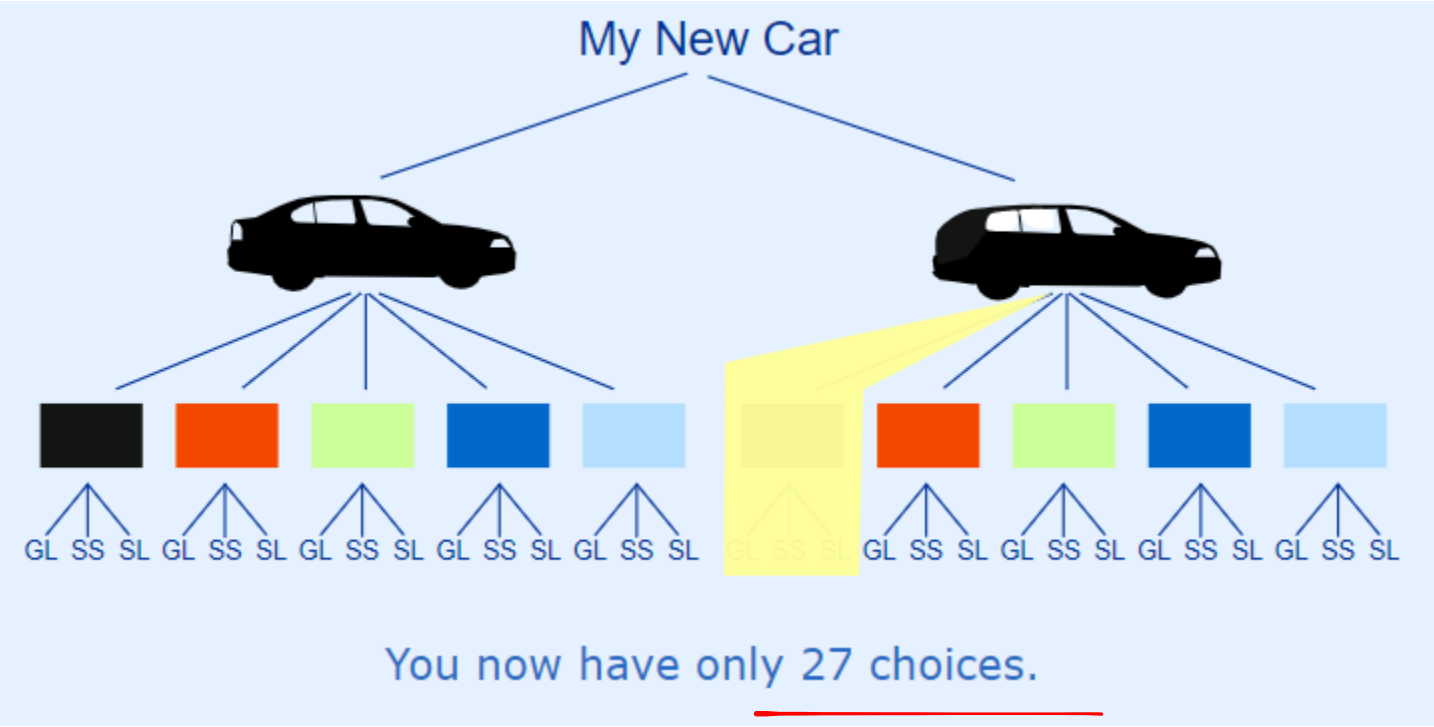


- You can count the choices, or just do the simple calculation:
- Total Choices = 2 x 5 x 3

Independent or Dependent?

- But it only works when all choices are **independent** of each other.
- If one choice affects another choice (i.e. **depends** on another choice), then a simple multiplication is not right.
- Example: You are buying a new car ... but ...
- the salesman says **"You can't choose black for the hatchback"** ... well then things change!

Independent or Dependent?



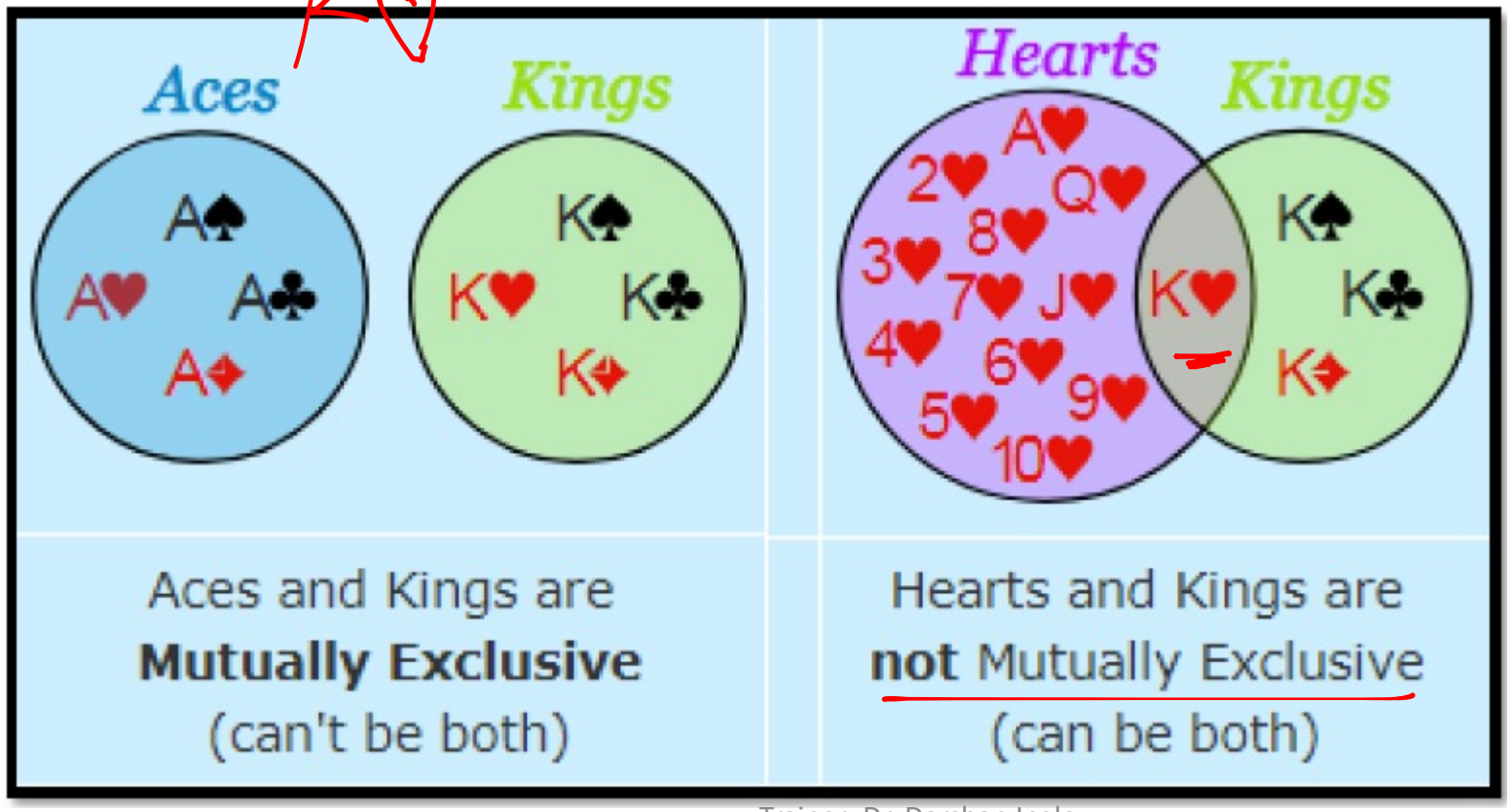
- Because your choices are **not independent** of each other.
- But you can still make your life easier with this calculation:
- Choices = $5 \times 3 + 4 \times 3 = 15 + 12 = 27$

Mutually Exclusive Events

- Mutually Exclusive means we can't get both events at the same time.
- Examples:

Mutually Exclusive Events

- What isn't Mutually Exclusive
  - Kings and Hearts are not Mutually Exclusive, because we can have a King of Hearts!



Mutually Exclusive Events

- When two events (call them "A" and "B") are Mutually Exclusive it is impossible for them to happen together:

P(A and B) = 0

"The probability of A and B together equals 0 (impossible)"

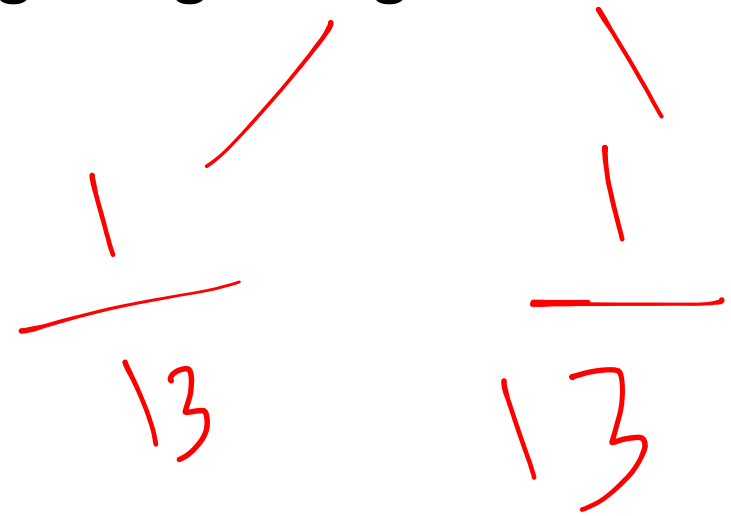
A card cannot be a King AND a Queen at the same time!

- The probability of a King **and** a Queen is **0** (Impossible)

But the probability of A **or** B is the sum of the individual probabilities:
P(A or B) = P(A) + P(B)

Mutually Exclusive Events

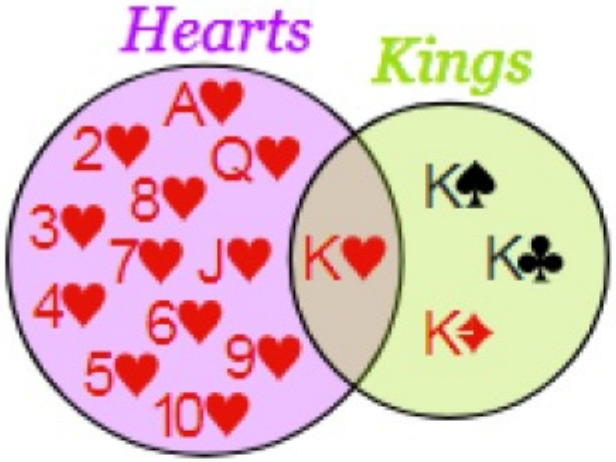
- Find what's the probability of getting a King OR Queen ?



$$P(K \text{ or } Q) = \frac{2}{13}$$

When events are not Mutually Exclusive ? - be careful in calculation !

What is the probability of getting a king or heart ?



But Hearts **or** Kings is:

- all the Hearts (13 of them)
- all the Kings (4 of them)

But that counts the King of Hearts twice!

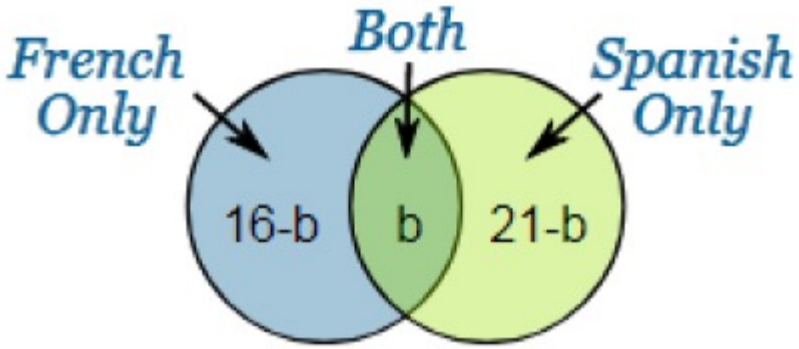
So we correct our answer, by subtracting the extra "and" part:



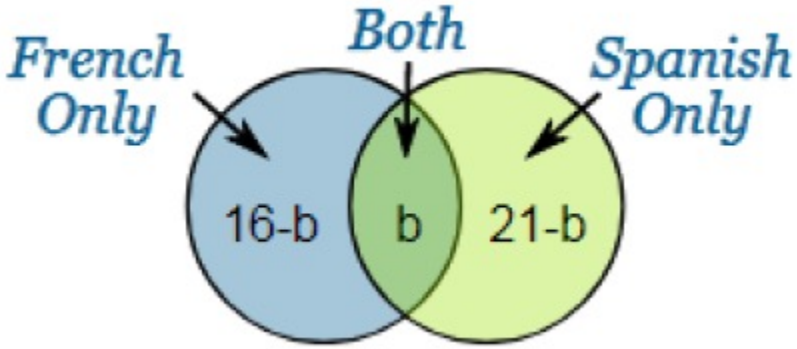
16 Cards = 13 Hearts + 4 Kings – the 1 extra King of Hearts

When events are not Mutually Exclusive ? - be careful in calculation !

- P(A or B) = P(A) + P(B) – P(A and B)
- Example: 16 people study French, 21 study Spanish and there are 30 altogether. Work out the probabilities!
- This is definitely a case of not Mutually Exclusive (you can study French AND Spanish).
- Let's say b is how many study both languages:
  - people studying French Only must be 16-b
  - people studying Spanish Only must be 21-b
- And we get:



When events are not Mutually Exclusive ? - be careful in calculation !



And we know there are **30** people, so:

- $(16-b) + b + (21-b) = 30$
- $37 - b = 30$
- $b = 7$

- P(French) = 16/30
- P(Spanish) = 21/30
- P(French Only) = 9/30
- P(Spanish Only) = 14/30
- P(French or Spanish) = 30/30 = 1
- P(French and Spanish) = 7/30

False Positives and False Negatives

When you have a test that can say "Yes" or "No" (such as a medical test), you have to think:

- It could be **wrong** when it says "Yes".
- It could be **wrong** when it says "No".

It is like being told you **did** something when you **didn't**!

Or you didn't do it when you really did.

False Positives and False Negatives

- They each have a special name: **"False Positive"** and **"False Negative"**:

	They say you **did**	They say you **didn't**
You really did	*They are right!*	**"False Negative"**
You really didn't	**"False Positive"**	*They are right!*

Some examples of "false positives" and "false negatives":

- **Airport Security:**
- **Quality Control:**
- **Antivirus software:**
- **Medical screening:**

Example

- Hunter says she is itchy. There is a test for Allergy to Cats, but this test is not always right:
  - For people that **really do** have the allergy, the test says "Yes" **80%** of the time
  - For people that **do not** have the allergy, the test says "Yes" **10%** of the time ("false positive")

	Test says "Yes"	Test says "No"
Have allergy	**80%**	**20% "False Negative"**
Don't have it	**10% "False Positive"**	**90%**

Question:

- If 1% of the population have the allergy, and Hunter's test says "Yes", what are the chances that Hunter really has the allergy?

- There are three different ways to solve this:
  - "Imagine a 1000",
  - "Tree Diagrams" or
  - "Bayes' Theorem"

Solution 1: Imagining A Thousand People

When trying to understand questions like this, just imagine a large group (say 1000) and play with the numbers:

- Of 1000 people, only **10** really have the allergy (1% of 1000 is 10)
- The test is 80% right for people who **have** the allergy, so it will get **8 of those 10 right**.
- But 990 **do not** have the allergy, and the test will say "Yes" to 10% of them, which is **99 people** it says "Yes" to **wrongly** (false positive)
- So out of 1000 people the test says "**Yes**" to (8+99) = **107 people**

	1% have it	Test says "Yes"	Test says "No"
Have allergy	10	**8**	**2**
Don't have it	990	**99**	**891**
	1000	107	893

Solution 1: Imagining A Thousand People

	1% have it	Test says "Yes"	Test says "No"
Have allergy	10	**8**	**2**
Don't have it	990	**99**	**891**
	1000	107	893

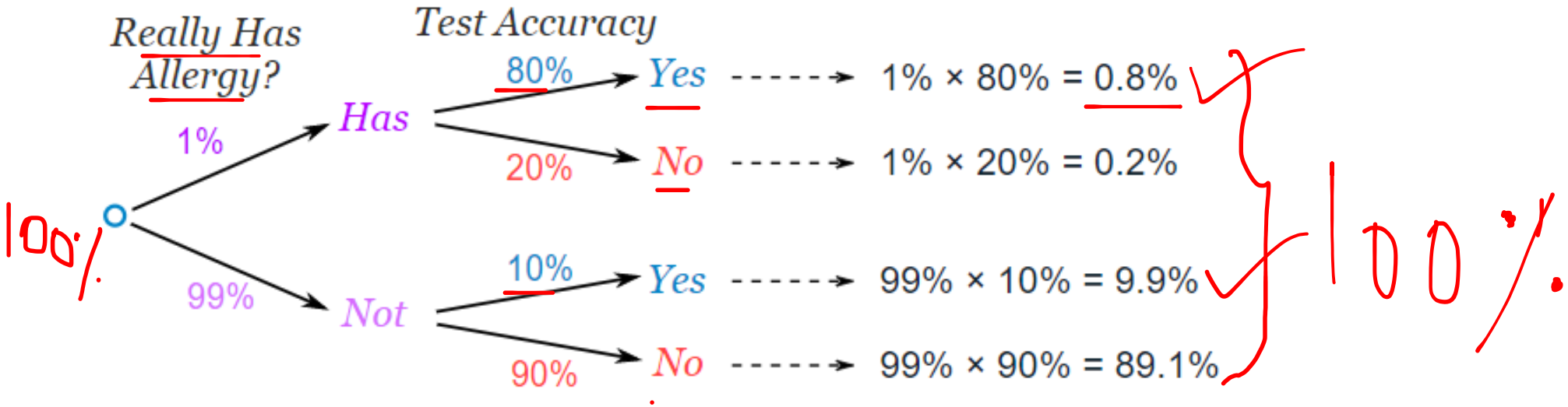
So 107 people get a "Yes" but only 8 of those really have the allergy:

8 / 107 = about 7%

So, even though Hunter's test said "Yes", it is still only **7% likely** that Hunter has a Cat Allergy.

Why so small? Well, the allergy is so rare that those who actually have it are greatly **outnumbered** by those with a false positive.

Solution 2: Tree Diagram



First of all, let's check that all the percentages add up:

0.8% + 0.2% + 9.9% + 89.1% = **100%** (good!)

And the two "Yes" answers add up to 0.8% + 9.9% = **10.7%**, but only 0.8% are correct.

0.8/10.7 = **7%** (same answer as above)

Solution 3: Bayes' Theorem

- Bayes' Theorem has a special formula for this kind of thing:

$$P(A|B) = \frac{P(A)P(B|A)}{P(A)P(B|A) + P(\text{not } A)P(B|\text{not } A)}$$

where:

- P means "Probability of"
- | means "given that"
- A in this case is "actually has the allergy"
- B in this case is "test says Yes"

Solution 3: Bayes' Theorem

To be clearer, let's change A to **has** (actually has allergy) and B to **Yes** (test says yes):

$$P(has|Yes) = \frac{P(has)P(Yes|has)}{P(has)P(Yes|has) + P(not\ has)P(Yes|not\ has)}$$

And put in the numbers:

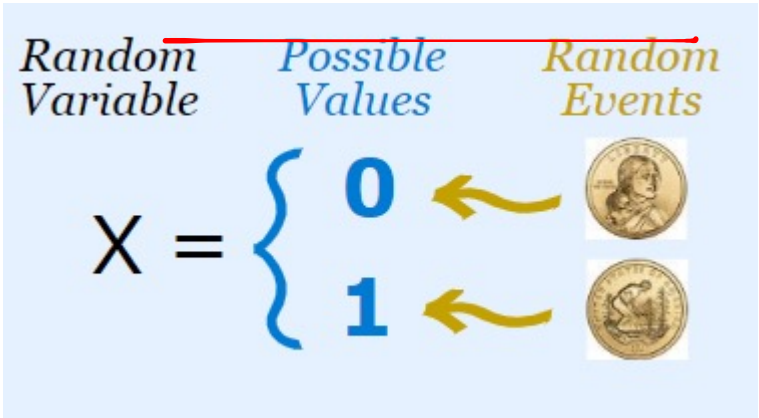
$$P(has|yes) = \frac{0.01 \times 0.8}{0.01 \times 0.8 + 0.99 \times 0.1}$$
$$= 0.0748...$$

Which is about **7%**

Random Variables



- A Random Variable is a set of **possible values** from a random experiment.
- Let's give them the values **Heads=0** and **Tails=1** and we have a Random Variable "X":



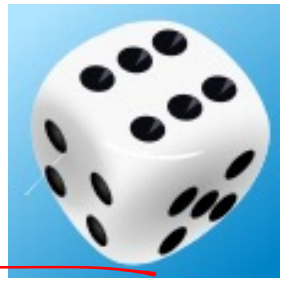
- In short:
  - X = {0, 1}

Random Variables

- **Not Like an Algebra Variable**
- In Algebra a variable, like **x,** is an unknown value:
- Example: x + 2 = 6
  - In this case we can find that x=4
- But a Random Variable is different ...
  - A Random Variable has a whole **set of values** and it could take on **any** of those values, randomly.
- Example: X = {0, 1, 2, 3}
  - X could be 0, 1, 2, or 3 **randomly**.
  - And they might each have a different probability.

Sample Space

- **A Random Variable's set of values is the Sample Space.**
- **Example:** Throw a die once
- Random Variable **X** = Score on Top Face



$X = \{1, 2, \ldots, 6\}$ $P(X) = \frac{1}{6}$

Probability

We can show the probability of any one value using this style:

$$P(X = \text{value}) = \text{probability of that value}$$

- Example (continued): Throw a die once

$X = \{1, 2, 3, 4, 5, 6\}$

In this case they are all equally likely, so the probability of any one is 1/6

- $P(X = 1) = 1/6$
- $P(X = 2) = 1/6$
- $P(X = 3) = 1/6$
- $P(X = 4) = 1/6$
- $P(X = 5) = 1/6$
- $P(X = 6) = 1/6$

Note that the sum of the probabilities = **1**, as it should be.

Example: How many heads when we toss 3 coins?



X = "The number of Heads" is the Random Variable.

In this case, there could be 0 Heads (if all the coins land Tails up), 1 Head, 2 Heads or 3 Heads.

So the Sample Space = {0, 1, 2, 3}

But this time the outcomes are NOT all equally likely.

The three coins can land in eight possible ways:

	X = "Number of Heads"
HHH	3
HHT	2
HTH	2
HTT	1
THH	2
THT	1
TTH	1
TTT	0 ✓

Looking at the table we see just 1 case of Three Heads, but 3 cases of Two Heads, 3 cases of One Head, and 1 case of Zero Heads. So:

- P(X = 3) = 1/8
- P(X = 2) = 3/8
- P(X = 1) = 3/8
- P(X = 0) = 1/8